

NED University of Engineering & Technology, Karachi

Discrete Structures (CT-162)

Mid Term Examination- 2024

Batch: 2024

Max Marks: 20

Time: 90 minutes

Date: October 9th, 2024

Instructions:

- In case of any ambiguity, you may make assumption. But your assumption should not contradict any statement in the question paper.
- Answering all the questions in given sequence of the question paper

Question # 1

(a) Let P, Q, and R be the propositions. (6 marks)

(CLO-2)

P: Niagara Falls is in New York. Q: New York City is the capital state of United State.

R: New York City will have more snow in 2050.

Write these propositions using P, Q, and R and logical connectives (including negations):

(i) If Niagara Falls is in New York, New York City will not have more snow in 2050.

(ii) Neither Niagara Falls is in New York nor will New York City have more snow in 2050.

(iii) It is not the case that New York City is not the state capital of the United States.

(iv) New York City will not have more snow in 2050 only if New York City is not the state capital of the United States.

(b) Using the truth table, prove or disprove that the contrapositive of statement (i) in part (a) is equivalent to the converse of its inverse.

Question #2

(4 marks)

(CLO-1)

Using Truth Table, show that these compound propositions are logically equivalent or not.

a. $(p \rightarrow r) \wedge (q \rightarrow r)$ and $(p \vee q) \rightarrow r$

b. $(p \rightarrow q) \rightarrow (r \rightarrow s)$ and $(p \rightarrow r) \rightarrow (q \rightarrow s)$

Question #3

(8 marks)

(CLO-3)

Prove the following equivalences by using laws of logic.

a. $(p \wedge q) \vee (\neg p \wedge q) \equiv q$

b. $\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg q$

c. $(p \wedge (\neg(\neg p \vee q))) \vee (p \wedge q) \equiv p$

d. $(\wedge q) \rightarrow (p \rightarrow q) \equiv T$

Question # 4

(2 marks)

(CLO-1)

Subtract $(10110)_2$ from $(11101)_2$ using 2's complement method.

$$\begin{array}{r} 01001 \\ - 10110 \\ \hline 01010 \end{array}$$

$$\begin{array}{r} 011101 \\ + 01010 \\ \hline 100111 \end{array}$$

$$1+0=1+1=0$$

$$\begin{array}{r} 10111 \end{array}$$